

Nicolás Barceló Lozano

AI & Software Engineer

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AI & Software Engineer with 26+ years of experience designing and implementing high-impact solutions across enterprise systems, document management, and process automation.

Currently focused on AI Agents, LLM orchestration, and intelligent automation, combining a strong foundation in .NET development with expertise in Python, NLP, and Generative AI.

Passionate about rapidly prototyping, testing, and scaling AI systems that enhance decision-making, optimize workflows, and deliver measurable business impact.

PROFESSIONAL EXPERIENCE

Tecnomedia Sistemas

Data and Documentation Technology Specialist

Barcelona, Spain

Nov 2002 – Present

AI Agents & Generative AI

- Built and deployed AI agents for decision automation and enterprise system integration using LangChain, LangGraph, CrewAI, n8n, and Azure AI.
- Integrated generative AI models through the OpenAI SDK, enabling rapid prototyping and reliable LLM-based automation.
- Designed multi-agent orchestration pipelines for real-world use cases (document workflows, healthcare scheduling, CRM integrations).
- Leveraging Agentic Programming principles and modern CLI tools (Claude Code, Gemini CLI, Antigravity) to build, test, and scale autonomous agent architectures and intelligent automation within company projects.

Document & Data Solutions

- Led development of document management applications in .NET (C# / WPF / WinForms) and SQL Server.
- Designed workflows for digitization, indexing, OCR/ICR, and barcode recognition in industrial and healthcare domains.

AI/ML Projects

- Developed content-based text classification (NLP) for knowledge management.
- Directed a quality prediction project in the chemical industry using ML models.

Integration & Automation

- Implemented RPA solutions with C#, Python, and low-code (n8n).
- Built REST APIs and integrated enterprise systems with Minimal API and FastAPI.

NexTReT

Programmer - Analyst Programmer

Barcelona, Spain

Dec 2000 – Nov 2002

- Developed enterprise applications in VB6, SQL Server, ASP (VBScript) for insurance and consultancy clients.
- Built HR and payroll management solutions with PowerBuilder.

Datasix Sistemas

Consultant

Barcelona, Spain

Oct 1999 – Dec 2000

- Designed and implemented Business Intelligence solutions (VBA, SQL Server, multidimensional analysis).
- Developed Data Warehouses and Data Marts and trained users in BI tools.

EDUCATION

IEBS Business School

Master's in AI, Data Science & ML and Big Data

Nov 2022 - Sept 2023

Universitat de Barcelona

Bachelor's Degree in Physics

Jun 1999

CERTIFICATIONS

- TensorFlow Developer Certificate (2024)
- AZ-900 Azure Fundamentals
- AI-900 Azure AI Fundamentals
- PCEP-30-02 – Certified Entry-Level Python Programmer
- PCAP-31-03 – Certified Associate in Python

SKILLS

AI & ML: LLMs, Agentic Programming, Prompt Engineering, AI Agents, LangChain, LangGraph, CrewAI, OpenAI SDK, Claude Code, Gemini CLI, Antigravity, NLP, RAG pipelines

Software Development: .NET (C#, VB.NET, WinForms, WPF), Python, JavaScript/TypeScript

Data & Cloud: SQL Server, NoSQL, Azure (Functions, AI, Synapse, Data Factory), Docker

Automation: RPA frameworks, n8n, API development (REST, OpenAPI)

BI & Visualization: Power BI, process mapping, human-in-the-loop systems

Languages: Native Spanish & Catalan · Professional English